



Bachelor's thesis in Physics

Formation and Evolution of Topological Defects in the Early Universe

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Abstract

This thesis provides an introduction to the physics and mathematics underlying the formation and evolution of topological defects. In the first half, after reviewing the expanding universe and classical field theory, we describe how spontaneous symmetry breaking in early Universe phase transitions may give rise to defect solutions in scalar field models. We focus on the case of domain walls and global strings and study their properties and dynamics from a mostly qualitative point of view.

In the second half of the thesis we discuss the numerical techniques used to simulate topological defects on a lattice. Using the open-source code *CosmoLattice*, we probe the dynamics of defect networks, focusing on features such as scaling behavior and the visualization of defects, and reproduce some results from the literature. Most notably, we modify *CosmoLattice*'s source code to extract the full complex field values at each lattice site, allowing us to visualize quantities such as the field's phase. We also analyze the dimensionless power-spectrum of a global string network and find that $\Delta^2 \sim (k/\mathcal{H})^{3.04}$ and $\Delta^2 \sim (k/\mathcal{H})^{-1.66}$ in IR and UV scales, respectively.

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0 Introduction

Topological defects are stable, localized structures that form when a physical system undergoes a phase transition and different regions settle into distinct, energy-minimizing configurations. Such phenomena appear in condensed matter systems when symmetries are spontaneously broken—for example, vortices in superfluid helium or domain walls in ferromagnets. In cosmology, similar mechanisms give rise to extended defects such as domain walls (2D) or cosmic strings (1D), depending on the nature of the symmetry and its breaking pattern. Zeldovich et al. [ZKO74] were the first to propose that such defects could have formed in the early universe during cosmological phase transitions, particularly those associated with grand unified theories (GUTs). A few years later, Tom Kibble provided a detailed description of their formation and introduced their classification via the homotopy groups of the vacuum manifold [Kib76].

Topological defects remain central in modern cosmology, as they are generic predictions of high-energy theories beyond the Standard Model involving spontaneous symmetry breaking. While certain defects like domain walls can lead to cosmological problems if overabundant, others, such as cosmic strings, remain viable and potentially observable. Understanding the scaling behavior of defect networks is essential for assessing their cosmological impact. Recent work has focused on high-resolution simulations of strings and walls, examining their scaling dynamics and the emission of axions or gravitational waves. These studies inform predictions of the axion relic abundance [HKS11; GHV21] and guide searches for stochastic gravitational wave backgrounds from strings and annihilating walls, see e.g. [Auc+20; NRT25], potentially detectable by experiments like LISA and pulsar timing arrays.

The first objective of this thesis is to develop a working understanding of the vast theory of topological defects, drawing from a broad range of topics in physics as well as mathematics. The second objective is to learn to use the *CosmoLattice* software in order to independently reproduce selected results from the literature on numerical simulations of topological defects and, where possible, to offer a novel perspective of the phenomena involved. We begin by reviewing the fundamental theory required to describe the simplest types of topological defects. This includes a brief overview of the expanding universe model and classical field theory, followed by the introduction of topological defects that arise from symmetry breaking in scalar field models under certain types of potentials, as presented in Chapter 1. Chapter 2 describes the lattice techniques used to simulate field theories in expanding backgrounds and Chapter 3 presents the results obtained for networks of domain walls and global strings.

1 The theory of topological defects

In this first chapter, we establish the foundational concepts relevant to the study of topological defects. We begin with a brief overview of the standard model of cosmology, which provides the physical context for the formation of such defects. We then introduce the essential elements of scalar field theory and discuss its generalization to curved spacetime. Following this, we present a broad classification of topological defects and the mechanisms underlying their formation. Finally, we focus in detail on domain walls and global cosmic strings which can be effectively described using scalar field models.

1.1 Preliminaries

There is a significant amount of theoretical background in both physics and mathematics required to describe topological defects. In this section our aim is to concisely present the essential concepts involved in the study of topological defect formation in the early universe. We assume the reader has a working knowledge of Special Relativity (SR), including its tensorial formulation. Throughout this thesis, we adopt the Einstein summation convention, i.e. $v^i u_i = \sum_i v^i u_i$, where Latin indices $i, j, \dots$ refer to spatial components and Greek indices $\mu, \nu, \dots$ refer to spacetime components. We use the $(-, +, +, +)$ signature and natural units $c = \hbar = G = 1$.

1.1.1 The standard model of cosmology

Before presenting the standard model of cosmology, we quickly review the theory of General Relativity, the theoretical framework on which it is built. The expansion of the universe, the formation of large-scale structures, and the evolution of its energy components all follow from the Einstein field equations formulated in this theory. A comprehensive treatment of GR can be found in [Car19], with Chapter 8 offering a quick introduction to the cosmological model¹.

The theory of *General Relativity (GR)* aims to generalize SR to allow for non-inertial/accelerated reference frames (RF) and to make it compatible with gravity. At the heart of GR is the *metric tensor* $g_{\mu\nu}$ which acts as the ruler of spacetime and is usually expressed in the components of the spacetime line element $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$. In SR $g_{\mu\nu} = \eta_{\mu\nu} \equiv \text{diag}(-1, 1, 1, 1)$ such that $ds^2 = -dt^2 + d\vec{x}^2$, which corresponds to the line element of flat/Minkowski spacetime in an inertial RF. In the presence of gravity or in a non-inertial RF the metric tensor deviates from $\eta_{\mu\nu}$ and this implies that spacetime is curved and free-falling objects move in non-straight paths through spacetime known as *geodesics*.

In Minkowski spacetime the partial derivative of a tensor $\partial_\alpha T^{\mu\nu\dots}_{\rho\lambda\dots}$ is well defined and preserves the tensorial nature of the object. However, in curved spacetime, the coordinates can vary from point to point in a non-trivial way, and taking partial derivatives does not account for the change in the basis vectors of the tangent space across the manifold. As a result, the partial derivative of a tensor does not transform as a tensor. This issue is overcome with the notion of the *covariant derivative* ∇_α which for a contravariant vector V^μ is defined as $\nabla_\alpha V^\mu = \partial_\alpha V^\mu + \Gamma^\mu_{\alpha\beta} V^\beta$; the terms involving the *Christoffel symbols* $\Gamma^\mu_{\alpha\beta}$ can be interpreted as the corrections to the partial derivative accounting for the curvature of the manifold.

In GR the *energy-momentum tensor* is the symmetric $(2, 0)$ -tensor $T^{\mu\nu}$ which contains the information of the distribution of energy and momentum in spacetime. $T^{\mu\nu}$ is conserved in the sense that its divergence vanishes $\nabla_\nu T^{\mu\nu} = 0$. Each component of the energy-momentum tensor has a physical interpretation: T^{00} represents the energy density, T^{0i} the i -th component of the momentum density and T^{ij} contains the terms related to pressure in the diagonal $i = j$ and shearing for $i \neq j$. In contrast to SR,

¹The treatment in this thesis does not include the cosmological constant Λ or cold dark matter.

in GR the energy-momentum tensor acts as the source of spacetime curvature. Gravity is not a force in the traditional sense but rather a manifestation of the curvature of spacetime caused by the presence of energy and momentum. This relationship is described by *Einstein's equations (EEs)*

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi T_{\mu\nu} \quad (1.1)$$

where $R_{\mu\nu}$ is the *Ricci curvature tensor* describing the curvature of spacetime and $R = g^{\mu\nu}R_{\mu\nu}$ is the *Ricci scalar*.

The *standard model of cosmology* is built upon the assumption that on sufficiently large scales (≥ 100 Mpc), the universe is homogeneous and isotropic. These symmetries constrain the possible forms of the metric tensor, leading to the *Friedmann-Lemaître-Robertson-Walker (FLRW)* metric

$$ds^2 = -dt^2 + a(t)^2 \left[\frac{dr^2}{1 - \kappa r^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right], \quad (1.2)$$

where $a(t)$ is the *scale factor* and κ encodes the spatial curvature: $\kappa > 0$ corresponds to a *spherical* universe, $\kappa = 0$ to a *flat* universe, and $\kappa < 0$ to an *hyperbolic* universe. The FLRW metric describes a spacetime that expands or contracts over time while preserving spatial homogeneity and isotropy. The time t is called *cosmic time* defined as the proper time measured by a *comoving observer*, i.e., an observer who is moving with the cosmic flow and the coordinates (r, θ, ϕ) are the (spherical) *comoving coordinates*.

The scale factor $a(t)$ describes how spatial distances evolve in an expanding universe. Comoving coordinates $\vec{x}$ remain fixed for observers at rest with the expansion, while physical distances grow as $a(t)\vec{x}$. The concept of a *horizon* arises naturally in this setting as it quantifies the maximal distance from which light could have reached an observer by a given time due to its finite speed and increase in distance it has to travel due to expansion. The *proper distance* to the particle horizon at t is given by

$$d_H(t) = a(t) \int_0^t \frac{dt'}{a(t')} \quad (1.3)$$

which leads naturally to the definition of *conformal time* η as $d\eta = dt/a(t)$ such that $d_H(\eta) = \eta a(\eta)$. In conformal time the FLRW metric takes the form

$$ds^2 = a(\eta)^2 \left[-d\eta^2 + \frac{dr^2}{1 - \kappa r^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right], \quad (1.4)$$

which for $\kappa = 0$ is just the Minkowski spacetime metric multiplied by the scaling factor. A natural observable derived from the scale factor is the *Hubble parameter*

$$H(t) \equiv \frac{\dot{a}(t)}{a(t)} \quad \text{or} \quad \mathcal{H}(\eta) = \frac{a'(\eta)}{a(\eta)} \quad (1.5)$$

where $\dot{a}$ denotes derivative w.r.t. cosmic time and a' w.r.t conformal time and $\mathcal{H} = aH$. The Hubble parameter quantifies the expansion rate of the universe at a given time, and, for our purposes, it also sets a characteristic scale for causal connectivity with $R_H(t) \equiv H^{-1}$ the *Hubble radius*. A *Hubble patch* is then defined as a region of the universe with a radius of the order $R_H(t)$.

The assumptions of the standard model imply that at large scales the matter content of the universe is adequately described as a perfect fluid, characterized by an *energy density* ρ and an *isotropic pressure*

p , such that the energy-momentum tensor has the form

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu + pg^{\mu\nu} \quad (1.6)$$

where u^μ is the 4-velocity of the fluid. The EEs impose constraints on the form of the scale factor, indeed substituting (1.6) in the right hand side of (1.1) and computing the left hand side for the FLRW metric (1.2) one finds the *Friedmann equations*

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3}\rho - \frac{\kappa}{a^2}, \quad \frac{\ddot{a}}{a} = -\frac{4\pi}{3}(\rho + 3p). \quad (1.7)$$

Differentiating the first equation in (1.7) with respect to time and substituting into the second leads to

$$\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + p) = 0. \quad (1.8)$$

Equation (1.8) is particularly important in determining how the energy density of the universe evolves as a function of the scale factor when combined with an *equation of state (EoS)*; that is, a relationship between the pressure and the energy density of the form $p = \omega\rho$ where ω is the *equation of state parameter*. If ω is constant we can plug the EoS into Eq. (1.8) and solve the resulting ODE to find that $\rho \propto a^{-3(1+\omega)}$ which can in turn be substituted into the first equation in (1.7) to obtain the following ODE for $a(t)$

$$\dot{a}^2 + \kappa \propto \frac{8\pi}{3}a^{-(1+3\omega)}. \quad (1.9)$$

No single value of ω describes the entire history of the universe but there are different periods of its life where there has been a predominant form of energy/matter. Each of these periods is known as an *epoch* of the universe and has a corresponding EoS. Briefly, in the standard model of cosmology the universe began in a hot, dense state, roughly 10^{-34} seconds after the Big Bang, where radiation dominated the energy density, causing rapid expansion. As the universe cooled matter took over, leading to the matter-dominated epoch, where structure formation began. Eventually, dark energy became the dominant component, causing the universe's expansion to accelerate in the dark energy-dominated epoch, a trend that continues today. Current measurements of the *cosmic microwave background (CMB)* suggest that the spatial flatness FLRW model gives the most accurate description of our universe [Agh+20]. For $\kappa = 0$ the ODE in Eq. (1.9) can be easily solved in terms of ω ; with the most relevant values and their corresponding epochs summarized in Table 1. It is worth noting that in a matter-dominated universe $\rho \propto a^{-3}$, which is expected since mass spreads out as volumes grow with a^3 . In contrast, during the radiation-dominated era $\rho \propto a^{-4}$, where the additional factor of a^{-1} is explained by accounting for the redshifting of photon energies as the universe expands.

Epoch	ω	$\rho(a) \propto$	$a(t) \propto$	$H(t)$	Time Period
Radiation	1/3	a^{-4}	$t^{1/2}$	$\frac{1}{2t}$	$10^{-34} \text{ s} \lesssim t \lesssim 50,000 \text{ yr}$
Matter	0	a^{-3}	$t^{2/3}$	$\frac{2}{3t}$	$50,000 \text{ yr} \lesssim t \lesssim 10 \text{ Gyr}$
Dark energy	-1	const.	e^{Ht}	const.	$t \gtrsim 10 \text{ Gyr}$

Table 1: Summary of the main cosmological epochs, characterized by the equation-of-state parameter ω , the scaling of the energy density $\rho(a)$, the scale factor $a(t)$, and the Hubble parameter $H(t)$, assuming spatial flatness ($\kappa = 0$).

1.1.2 Classical field theory

In modern physics, fields are the primary language for understanding fundamental particles and their interactions. In particular, classical field theory provides the tools for describing topological defects such as domain walls and cosmic strings. This section briefly introduces the necessary formalism, focusing on scalar and complex fields in flat and curved spacetime. For a detailed treatment of classical field theory see [Näs19].

A *field* is a quantity defined at every point depending smoothly on the coordinates. *Scalar fields* are real valued functions of the coordinates $\phi = \phi(x^\mu)$ and *complex fields* consist of a complex function whose real and imaginary components are scalar fields. Like in particle classical mechanics, the dynamics of a field are encoded in the *Lagrangian density* $\mathcal{L}(x^\mu, \phi, \partial_\mu \phi)$ which for a free scalar field in Minkowski spacetime usually adopts the form

$$\mathcal{L} = -\frac{1}{2}\partial_\mu \phi \partial^\mu \phi - V(\phi), \quad (1.10)$$

where $V(\phi)$ is the *potential energy*. The canonical example of a potential for a scalar field is the *harmonic potential* $V(\phi) = \frac{1}{2}m^2\phi^2$ where m is a parameter known as the *field's mass*. The *action* S is the spacetime integral of the Lagrangian density and extremizing it gives the field's *Euler-Lagrange (EL)* equation

$$S[\phi] = \int d^4x \mathcal{L}(x^\mu, \phi, \partial_\mu \phi) \stackrel{\delta S=0}{\implies} \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) = 0; \quad (1.11)$$

the solution to EL equation describes the dynamics of ϕ . For the Lagrangian of Eq. (1.10) the resulting EL equation is

$$\square \phi - \frac{dV}{d\phi} = 0 \quad (1.12)$$

where $\square \equiv \partial_\mu \partial^\mu = -\partial_{tt} + \vec{\nabla}^2$ is the *d'Alembert operator*. In the particular case of the harmonic potential Eq. (1.12) is known as the *Klein-Gordon (KG) equation* and for $m = 0$ it is the wave equation. We note that the harmonic potential has a single minimum at $\phi = 0$. For a general potential $V(\phi)$, field configurations that minimize $V(\phi)$ are called *vacuum states*. Around a vacuum state, a potential $V(\phi)$ can be expanded in a power series to have the form of a harmonic potential up to $O(\phi^2)$. The mass of a field around the vacuum ϕ_0 is defined as

$$m^2 \equiv \left. \frac{d^2 V}{d\phi^2} \right|_{\phi=\phi_0} \quad (1.13)$$

such that around a vacuum a field will solve the corresponding KG equation. A potential with *degenerate vacua* is one having multiple minima of the same value; the set of all vacuum states of a potential is known as the *vacuum manifold* and is denoted by $\mathcal{M}$.

To generalize the previous ideas to a curved spacetime we have to change the Minkowski metric for the metric tensor $\eta_{\mu\nu} \mapsto g_{\mu\nu}$ and partial derivatives for covariant derivatives $\partial_\mu \mapsto \nabla_\mu$. Additionally, when integrating over curved spacetime we have to take into account how the curvature of spacetime affects volumes $d^4x \mapsto \sqrt{-g} d^4x$, where $\sqrt{-g} \equiv \sqrt{-\det g_{\mu\nu}}$. With these modifications we obtain the action in curved spacetime for a scalar field

$$S[\phi] = \int d^4x \sqrt{-g} \mathcal{L}(x^\mu, \phi, \nabla_\mu \phi), \quad \mathcal{L} = -\frac{1}{2}\nabla_\mu \phi \nabla^\mu \phi - V(\phi) \quad (1.14)$$

and the EL equation in terms of the curved spacetime d’Alambertian

$$\square\phi - \frac{dV}{d\phi} = 0, \quad \square \equiv \nabla_\mu \nabla^\mu = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu). \quad (1.15)$$

For example, consider the metric of the FLRW model in cosmic time (Eq. (1.2)) with $\kappa = 0$ then Eq. (1.15) becomes

$$\ddot{\phi} + 3H\dot{\phi} - \frac{1}{a^2} \vec{\nabla}^2 \phi + \frac{dV}{d\phi} = 0. \quad (1.16)$$

In the case of a harmonic potential, Eq. (1.16) takes the form of a damped harmonic oscillator, where $3H\dot{\phi}$ acts as a friction term. This term is referred to as *Hubble friction*, as it arises from the expansion of the universe and damps the evolution of the field. The field is said to begin *rolling* once the potential starts dominating over Hubble friction—that is, when $H \lesssim m$. It is often more convenient to work in conformal time due to the scale factor in the spatial Laplacian in Eq. (1.16) breaking the wave like nature of the PDE. In conformal time the EoM is

$$\phi'' + 2\mathcal{H}\phi' - \vec{\nabla}^2 \phi + a^2 \frac{dV}{d\phi} = 0 \quad (1.17)$$

which is more easily interpreted as a wave/KG type equation with a time dependent potential.

1.2 Topological defects

Topological defects are stable, localized field configurations that emerge when a system undergoes spontaneous symmetry breaking and the resulting vacuum manifold has non-trivial topology. Their formation in the early universe is explained by the Kibble-Zurek mechanism, which describes how causally disconnected regions can fall into different vacuum states during phase transitions. Topological defects are classified by their dimensionality using homotopy theory, depending on the topology of the vacuum manifold. These objects are of interest because they are generic predictions of many GUTs and appear in certain models of dark matter. For a detailed exposition, see [VS00].

1.2.1 Classification and formation of topological defects

The first step in understanding the formation of topological defects is analyzing how the symmetries of a theory break down. A symmetry is said to be *global* if it does not depend on the spacetime point x^μ . While a given Lagrangian may possess a global symmetry it is possible for the vacuum of the theory to break this symmetry. This phenomenon is known as *spontaneous symmetry breaking (SSB)*. Formally, let G be the global symmetry group of the action and let ϕ_0 denote a vacuum state of the field. If there exists a transformation $g \in G$ such that $g \cdot \phi_0 \neq \phi_0$, then the vacuum does not respect the symmetry, even though the action is invariant. In this case, the symmetry is said to be spontaneously broken. According to the *Goldstone theorem* [Nae21], if a continuous global symmetry is spontaneously broken, there exists a *massless scalar field*, known as a *Goldstone boson*, for each generator of the symmetry group that does not leave the vacuum invariant. These correspond to excitations along the vacuum manifold $\mathcal{M}$.

The topological structure of $\mathcal{M}$ plays a central role in the formation of topological defects. In particular, the possible types of defects are classified by the *homotopy groups* $\pi_n(\mathcal{M})$, see Table 2. Roughly speaking, $\pi_n(\mathcal{M})$ is the set of equivalence classes of continuous maps from the n -dimensional sphere S^n to $\mathcal{M}$, where two maps are equivalent if one can be continuously deformed into the other. The group $\pi_n(\mathcal{M})$ is said to be *trivial* if all such maps are continuously deformable into each other.

Defect Type	Non-trivial Homotopy Group	Spatial Codimension
Domain Wall	$\pi_0(\mathcal{M})$	1
Cosmic String	$\pi_1(\mathcal{M})$	2
Monopole	$\pi_2(\mathcal{M})$	3

Table 2: Classification of topological defects by homotopy group and codimension.

To illustrate these concepts, consider the simplest example of a potential admitting a continuous symmetry involving a complex scalar field φ

$$V(\varphi) = \frac{\lambda}{4} (|\varphi|^2 - v^2)^2, \quad (1.18)$$

where $\lambda, v > 0$. The set of field configurations minimizing V is given by $\varphi = ve^{i\theta}$ with $\theta \in [0, 2\pi)$, which defines a vacuum manifold topologically equivalent to the circle S^1 . The potential is invariant under the global $U(1)$ symmetry $\varphi \mapsto e^{i\alpha}\varphi$ where α is any real constant. However, once a specific vacuum configuration $\varphi_0 = ve^{i\theta_0}$ is chosen, this symmetry is spontaneously broken, as $\varphi_0 \neq ve^{i(\theta_0+\alpha)}$ for generic α . The Goldstone boson in this case corresponds to excitations along the angular direction of the circle defined by $|\varphi| = v$. It can be shown that the only non-trivial homotopy group of the circle corresponds to $n = 1$ such that $V(\varphi)$ has a single Goldstone boson and the corresponding topological defect is a string.

The *Kibble-Zurek mechanism* describes how different regions of space can independently fall into different vacuum states during phase transition of the universe. As these regions come into causal contact, their mismatched field configurations can result in topologically stable structures at their boundaries. In order to explain this process and the appearance of the potential in Eq. (1.18), it is necessary to take into account the thermal environment of the early universe. At high temperatures, thermal corrections to the effective potential modify the shape of the vacuum structure. For the potential in Eq. (1.18) a typical form of the temperature-dependent effective potential during a phase transition is

$$V(\varphi, T) = \frac{\lambda}{4} (|\varphi|^2 - v^2)^2 + \frac{1}{2} c T^2 |\varphi|^2, \quad (1.19)$$

where T is the temperature and $c > 0$ is a constant that depends on the details of the field's interactions. At sufficiently high temperatures $T \gg T_c$, the thermal mass term $\sim T^2 |\varphi|^2$ dominates and the potential is KG-like with a single minimum at $\varphi = 0$. As the universe expands and cools, the temperature drops below a *critical value* T_c , at which point the potential develops a ring of degenerate minima close to $|\varphi| = v$. This signals the spontaneous breaking of the symmetry given the non-trivial vacuum manifold $\mathcal{M}$. However, the process of symmetry breaking does not occur uniformly throughout space due to the finite speed at which information can propagate. During the phase transition, a vacuum is randomly chosen at each point in space, influenced by thermal fluctuations of the field, such that two different points can only be correlated if they are separated a length of at most R_H . Consequently, each Hubble patch randomly selects a different vacuum configuration which may give rise to defects in the patch boundaries once they eventually come into contact.

1.2.2 Domain walls

The simplest model that allows for spontaneous symmetry breaking is one with a discrete $\mathbb{Z}_2$ symmetry. An example can be constructed from a real scalar field ϕ under the potential

$$V(\phi) = \frac{\lambda}{4} (\phi^2 - v^2)^2. \quad (1.20)$$

This potential has two degenerate minima at $\phi = \pm v$, which constitute the vacuum manifold $\mathcal{M}$. The $\mathbb{Z}_2$ symmetry refers to the invariance of the potential under the transformation $\phi \mapsto -\phi$. Since this is a discrete symmetry, its spontaneous breaking does not lead to a Goldstone boson. Nonetheless, the vacuum structure of the theory admits the formation of topological defects known as *domain walls*. In particular, the vacuum manifold is disconnected implying that $\pi_0(\mathcal{M})$ is not trivial.

To understand the structure of domain walls, we consider the simplest case: a static planar wall in flat 1 + 1 Minkowski spacetime. The EoM derived from the action is

$$\phi''(x) - \lambda\phi(\phi^2 - v^2) = 0 \quad (1.21)$$

which admits the non-trivial analytic solution

$$\phi(x) = v \tanh\left((\lambda/2)^{1/2}vx\right) \quad (1.22)$$

satisfying the boundary conditions $\phi = -v$ as $x \rightarrow -\infty$ and $\phi = +v$ as $x \rightarrow +\infty$. The domain wall is localized in a region of width $\delta_w \sim (\lambda^{1/2}v)^{-1}$ and centered around $x = 0$, where the field changes rapidly and the potential energy is concentrated. Moreover, the mass of the field is $m_\phi = \lambda^{1/2}v$ so the wall width is determined by the mass $\delta_w \sim m_\phi^{-1}$.

The stability of the domain wall configuration is topological. Because the vacuum manifold consists of two disconnected points, any field configuration interpolating between them cannot be continuously deformed into a trivial configuration without crossing the potential barrier, which would require infinite energy. This solution is also *localized*, meaning its total energy is located in a bounded region of space and it is given by

$$E(\phi) = \int dx \left[\frac{1}{2}\phi'(x)^2 + V(\phi) \right] = \frac{2\sqrt{2}}{3}\lambda^{1/2}v^3, \quad (1.23)$$

which is finite. In 1 + 3 Minkowski spacetime the same solution can be found by further assuming that the wall lies in the yz -plane. However, this solution is not localized as it spans a whole plane of space and its energy diverges. This is actually a consequence of Derrick's theorem [Näs19] stating that there are not stable and time-independent localized solutions for such scalar models in dimension $D > 1$. The energy momentum tensor of the wall is

$$T_\nu^\mu = \phi'(x)^2 \text{diag}(1, 0, 1, 1); \quad (1.24)$$

in particular the surface energy density of the wall is $\sigma \sim \lambda^{1/2}v^3$ which coincides with the wall tension in the two tangential directions due to the time and y and z components of T_ν^μ being equal.

Although the previous model does not take into account an expanding universe, it is useful to explain the basic properties and behaviors of domain walls in a more general setting. An analytical treatment of the evolution of domain walls is complex and has not been yet fully developed so we will turn to numerical simulations to probe the dynamics of domain walls. Qualitatively, from the Kibble mechanism it is expected that following the symmetry breaking different regions of the universe choose either v or $-v$ and as these regions come in to contact domain walls form at their interfaces. The result will be a *network* of domain walls whose shape is irregular in general unlike the flat wall solution in Eq. (1.22). Locally, the domain walls will resemble the flat solution whenever their radius of curvature is much larger than the spatial scale.

In a first approach, due to their two-dimensional nature, it is expected that the energy density of a domain wall network scales as $\rho_{DW} \sim a^{-1}$. In a universe comprised only of domain walls, which we

treat as a fluid, this would correspond to an equation of state parameter $\omega = -2/3$. However, this is only a rough estimate, as it doesn't take into account interactions between walls and energy loss in the form of scalar and gravitational radiation. It is predicted by the theory [VS00] and has been observed in simulations, e.g. [LM11], that, after forming, domain wall networks quickly settle into what's known as the *scaling regime*. In this regime, the number of walls per Hubble patch stays roughly constant, and the network evolves in a self-similar way. This is due to the fact that domain walls can collide, reconnect, and form closed surfaces which then collapse under their own tension, radiating energy into scalar waves. As a consequence, the energy density of the walls scales as $\rho_{DW} \sim \sigma H$, so during radiation domination, where $H \propto a^{-2}$, one finds $\rho_{DW} \sim a^{-2}$. During matter domination the scaling becomes slower, $\rho_{DW} \sim a^{-3/2}$, but, in both cases, the domain wall energy dilutes more slowly than the background. This means that wall interactions in the network add to the down-scaling of energy density but more importantly it also implies that domain walls would eventually dominate the energy of the universe, leading to large-scale anisotropies unless their tension is small enough to evade observational bounds [ZKO74]. As a consequence, there cannot be any stable, highly-energetic domain walls in our universe today; this is known as the *domain wall problem*. However, several models predict the formation of domain walls in the early universe, which could have subsequently decayed if one of the vacua is favored over the others [LSW97]; see [Fer+24] for a recent numerical study. It has also been proposed that the annihilation of domain walls in the early universe could have sourced a current background of gravitational waves [HKS11] and shown numerically, e.g. [NRT25].

1.2.3 Global strings

Cosmic strings are one-dimensional, string-like topological defects which arise from the breaking of a $U(1)$ symmetry. A complex field φ under the potential in Eq. (1.18) is an example of a model which admits spontaneous symmetry breaking of the global $U(1)$ symmetry $\varphi \mapsto e^{i\alpha}\varphi$, leading to the formation of *global strings*. One can also consider models with *local* $U(1)$ symmetry of the form $\varphi \mapsto e^{i\alpha(x)}\varphi$, where now $\alpha(x)$ depends on the point in spacetime. The breaking of such a symmetry gives rise to *local* or *gauge* strings. Due to the interactions between the complex field and the gauge field, local strings experience *Higgs screening*, which cancels the long-range interactions of the field, resulting in the string energy per unit length being finite and localized near the string core. In this work, we will focus on the case of global strings, since their mathematical description and numerical simulation are simpler.

As in the case of domain walls, the general structure of global strings can be understood by studying a simpler model. Consider the potential in Eq. (1.18) for a complex field in $1+2$ Minkowski spacetime, the resulting EoM is

$$-\partial_{tt}\varphi + \partial_{xx}\varphi + \partial_{yy}\varphi - \frac{\lambda}{2}(|\varphi|^2 - v^2)\varphi = 0. \quad (1.25)$$

Assuming the solution is static, we propose the ansatz $\varphi(r, \theta) = e^{i\theta}f(r)$ in polar coordinates. We impose the boundary conditions $f(r) \rightarrow v$ as $r \rightarrow \infty$ and $f(0) = 0$ for the field to be continuous and a non-trivial solution. This yields the following ODE for f

$$\frac{d^2f}{dr^2} + \frac{1}{r}\frac{df}{dr} - \frac{1}{r^2}f - \frac{\lambda}{2}f(f^2 - v^2) = 0, \quad (1.26)$$

which has the asymptotic solutions

$$f(r) \approx cr \quad \text{for } r \ll v, \quad f(r) \approx v - \mathcal{O}(r^{-2}) \quad \text{for } r \gg v. \quad (1.27)$$

We see that close to the origin $\varphi \approx 0$, implying that this is a region of high energy, which we identify

as the string core. As in the case of domain walls, the width of the string core is related to the inverse mass of the field, $\delta_s \sim m_\varphi^{-1} = (\lambda^{1/2}v)^{-1}$. This solution can then be extended along the z -axis to obtain a static, cylindrically symmetric string with energy per unit length μ , i.e., the *string tension*. We can approximate μ by integrating the energy density in the transverse plane for $r \gg \delta_s$ and up to a cutoff distance λ ,

$$\mu = \int d^2x \left[\frac{1}{2}(\vec{\nabla}\varphi)^2 + V(\varphi) \right] \approx \int_0^{2\pi} d\theta \int_{\delta_s}^{\lambda} r \frac{v^2}{2r^2} dr \sim v^2 \log \left(\frac{\lambda}{\delta_s} \right). \quad (1.28)$$

Unlike the case of local strings, the tension of a global string diverges logarithmically, which is why we have introduced the cutoff. In practice, this cutoff corresponds to the largest physical scale over which the string extends. This could be the Hubble radius, or alternatively, the distance to the nearest string in the network.

As mentioned before, the vacuum manifold of the string potential is topologically equivalent to the circle S^1 having non-trivial $\pi_1(\mathcal{M})$. We can use this fact to understand the geometry of string defects from the Kibble–Zurek mechanism. Indeed, if at early times in the universe different causally disconnected regions of spacetime have fallen into one of the vacua corresponding to some phase θ , then as they come into causal contact, the field interpolating between them must be continuous. If it so happens that the regions coming into contact differ by a non-zero phase in their corresponding vacua within a given volume, there must be a point in between them where the field must vanish, otherwise the field would not be continuous.

Like domain walls, string formation occurs in the form of networks, which lead to string interactions that favor annihilation and thus contribute to energy dilution. If one considers only the effects of the expanding background, the energy density of a string network is expected to scale as $\rho \sim a^{-2}$, due to their one-dimensional nature. However, it is known that strings interact in a highly efficient way, which further increases the rate of energy dilution [VS00]. A particularly interesting consequence of these interactions is the formation of string loops, whose decay may result in axion emission [BS94] (in the case of global strings). *Axions* are hypothetical elementary particles that arise as the Goldstone bosons associated with the spontaneous breaking of a global $U(1)$ symmetry, such as those connected to global cosmic strings. It has been theorized that axions may constitute a significant component of the cold dark matter inferred from current cosmological observations [HN09]. String networks also tend to evolve into a scaling regime in which their energy density scales as $\rho_S \sim \mu H^2$ with the average number of strings per Hubble patch being constant. If this scaling behavior sets in during radiation domination, then $\rho_S \sim a^{-4}$, implying that strings dilute at the same rate as the rest of the radiation content of the universe. This also holds during matter domination. As a result, the possibility that our current universe contains cosmic strings has not been excluded and may be observable via gravitational lensing effects. However, current observations have placed upper bounds on the energy of such defects [BM10].

2 Lattice simulations of topological defects

As hinted in Chapter 1, the partial differential equations that arise from models of topological defects are generally very difficult to solve analytically. Over the past several decades, numerical methods have proven highly effective in studying the behavior of topological defects. This approach has significantly improved our understanding of their dynamics, the constraints associated with their formation and energy, and has helped determine how such defects might be observed. In this chapter, our goal is to describe the method and tools we will be using to simulate topological defects on a lattice, particularly for domain walls and global strings.

2.1 Simulating field theories with *CosmoLattice*

To simulate topological defects, we use the open-source software *CosmoLattice*, a comprehensive and flexible program written in C++ for lattice simulations of scalar and gauge field theories in expanding cosmological backgrounds. *CosmoLattice* functions as a sandbox where the lattice and the field evolution algorithms are already implemented with the user having to define the relevant aspects of the model such as the potential energy, parameters and initial conditions. For complete documentation of the program, see [Fig+23], and for a detailed explanation of the mathematical framework behind lattice simulations, see [Fig+21].

The basic idea behind lattice simulations is to discretize three-dimensional space into a grid of points within a finite cubic box. This box has a fixed side length L , which limits the spatial extent of the simulation. To mitigate this limitation and emulate an effectively infinite space, periodic boundary conditions are imposed; identifying opposite faces of the box and eliminating artificial edge effects. Each spatial dimension of the box is divided into N lattice points, such that the spacing between neighboring points is $\delta x = L/N$. The full lattice thus contains N^3 grid points, located at positions

$$\mathbf{x} = \delta x \mathbf{n}, \quad \text{with} \quad \mathbf{n} = (n_1, n_2, n_3), \quad n_i = 0, 1, \dots, N-1 \quad \text{for } i = 1, 2, 3. \quad (2.1)$$

Fields are then represented by their values at each lattice site, and their spatial and temporal derivatives are approximated using finite differences. In the lattice, *CosmoLattice* works with a particular set of dimensionless field and spacetime variables, denoted by $\{\tilde{\phi}, \tilde{\eta}, \tilde{x}^i\}$, which are referred to as *program variables*. These are related to the physical variables via the following transformations

$$\tilde{\phi} \equiv \frac{\phi}{f_*}, \quad d\tilde{\eta} \equiv a^{-\alpha} \omega_* d\tau, \quad d\tilde{x}^i \equiv \omega_* dx^i, \quad (2.2)$$

where f_* and ω_* are constants with dimensions of energy². When setting up a simulation, the choice of f_* and ω_* is made such that the resulting dimensionless variables are typically of order unity to simplify the analysis of the results. A common choice is to take f_* as the initial amplitude of the field and ω_* as its initial oscillation frequency. The choice of time variable is also important. While in this thesis we will only consider cosmic and conformal time, in principle an infinite number of time parametrizations are possible. A general rule for selecting a time variable is to keep the oscillation frequency of the field approximately constant. For scalar fields, this depends on the highest power of ϕ appearing in the potential. In particular, if $V(\phi) \propto \phi^4$, conformal time is the most appropriate choice (Eq. (35) in [Fig+23]).

An expanding universe introduces an additional challenge for its lattice representation: the lattice becomes dynamical. To address this, lattice points are identified with comoving coordinates so that the

²Here $\alpha = 0$ corresponds to cosmic time and $\alpha = 1$ to conformal time, τ is a generic time parameter.

box maintains a constant comoving width. As a result, non-expanding objects within the box will appear to shrink over time, even though their physical size remains constant. *CosmoLattice* also supports self-consistent evolution of the cosmological background. This means the effects of the field on the expansion of the universe are taken into account by solving the Friedmann equations alongside the field equations. However, in this work, we will not use this functionality. Instead, we will study the evolution of defects in fixed backgrounds, particularly radiation and matter domination such that a is predetermined. We will assume that both time and Hubble parameters are normalized to unity at the initial time, which we take without loss of generality to be $\eta = 0$. This sets the initial conditions $a(\eta = 0) = 1$, $\mathcal{H}(\eta = 0) = 1$. A general power-law expansion in conformal time, $a \sim \eta^p$ with $p = 2/(1 + 3\omega)$, yields

$$a(\eta) = \left(1 + \frac{\eta}{p}\right)^p \quad \text{and} \quad \mathcal{H}(\eta) = \frac{1}{1 + \frac{\eta}{p}}. \quad (2.3)$$

By default, *CosmoLattice* initializes all fields with a given set of deterministic initial conditions with an added stochastic fluctuation meant to mimic quantum vacuum fluctuations. In practice this corresponds to drawing the fluctuations of a field $\delta\tilde{\phi}(\tilde{\mathbf{n}})$ and its derivative $\delta\tilde{\phi}'(\tilde{\mathbf{n}})$ as Gaussian random fields. These fluctuations arise from the quantum nature of fields and generally depend on both the mass of the field and the expansion of the universe.

To simulate the actual dynamics of field theories, *CosmoLattice* provides several numerical integration schemes. Without delving into the full technical details, we will primarily use second-order methods, offering a good compromise between computational speed and accuracy. In particular, we will make use of the lattice formulations of the Velocity Verlet and Leapfrog algorithms. For the detailed explanation on how these methods are implemented see chapter 4 [Fig+21]. One final important detail to keep in mind when running lattice simulations is the Courant-Friedrichs-Lewy (CFL) condition, which imposes a constraint that must be satisfied to ensure numerical stability [LeV07]. In our case, the condition takes the form

$$\delta\tau \leq \frac{L}{N\sqrt{3}}. \quad (2.4)$$

It is important to note that in this form the condition is necessary but not sufficient: in practice, a smaller time step is often required to avoid instabilities. In general, increasing the lattice resolution $\sim (\delta x)^{-1}$ demands both a larger number of lattice points N and a finer time step $\delta\tau$. This significantly increases computational cost, with system memory quickly becoming a limiting factor.

2.2 Simulating domain walls

To correctly capture the formation and dynamics of domain walls under the potential of Eq. (1.20), certain general conditions must be met that constrain the values of the parameters involved in the simulation. On one hand, it is important to ensure that domain walls do not become thinner than the physical resolution of the lattice due to the cosmic expansion. Comparing the physical width of a wall δ_w to the physical lattice spacing $\delta x_{\text{phys}} = a\delta x$, the condition that walls remain resolvable becomes is $\delta_w \gg \delta x_{\text{phys}}$. Approximating δ_w by its value for planar walls we find $(\sqrt{\lambda}v)^{-1} \gg aL/N$. In the power-law expansion of Eq. (2.3), this sets a maximum time beyond which domain walls are no longer adequately resolved

$$\eta_{\text{max}} \approx p \left[\left(\frac{N}{Lv\sqrt{\lambda}} \right)^{\frac{1}{p}} - 1 \right]. \quad (2.5)$$

On the other hand, it is also essential that the simulation box contains a sufficiently large number of Hubble patches at the time of domain wall formation. This ensures that the Kibble-Zurek mechanism results in a statistically homogeneous network of domain walls. Otherwise, if the number of Hubble

patches is too small, the resulting field configuration may be dominated by a single vacuum, potentially suppressing domain wall formation altogether. The condition for this is that the physical box size exceeds the Hubble radius $aL \gg H^{-1}$, or equivalently $L \gg \mathcal{H}^{-1}$. This sets a second maximum time

$$\eta_{\max} \approx p(L - 1). \quad (2.6)$$

Equating conditions (2.5) and (2.6) yields an optimal relationship between the parameters, maximizing the duration for which the simulation remains accurate

$$N = L^{p+1} v \sqrt{\lambda}. \quad (2.7)$$

We note, however, that near the limiting time the simulation results may already begin to lose accuracy due to a reduced number of Hubble patches and limited spatial resolution. Furthermore, the wall width estimate used above corresponds to that of a static flat wall and may not fully capture the dynamical behavior.

As for the choice of parameters, we aim for dimensionless simulation quantities to be of order unity to aid interpretability, so we set $v = 1$. The mass of the field under $V(\phi)$ given by Eq. (1.13) is $m = \sqrt{2\lambda}$. Because we initialize with $\mathcal{H} = 1$, the condition for rolling motion to begin promptly implies $\lambda = 1/2$ is an appropriate choice. Setting $N = 256$ we find that for radiation domination ($p = 1$), $L \approx 19$. This yields approximately 6000 Hubble patches at the start and a maximum conformal time of $\eta_{\max} \approx 18$. In Appendix C.2 we include simulations for a matter-dominated background, where the discussion on the choice of parameters is also found. As for the initial conditions, we set both the field and its momentum to zero, so that the only effective initial input comes from the random fluctuations. It is worth noting that the conditions under which topological defects form typically involve thermal fluctuations, rather than the quantum vacuum fluctuations `CosmoLattice` incorporates. While the two are not the same, the main difference lies in the amplitude of the fluctuations. This difference can be ignored as long as the amplitudes are not orders of magnitude apart. In practice, this means that the initial field configuration should not be too energetic. Fluctuations that are too small are not necessarily problematic, since the field will eventually undergo symmetry breaking; however, they will delay the formation of domain walls. This can be numerically inconvenient, as a late formation time may lead to a smaller number of Hubble patches in the box by the time the defects appear.

Finally, using `CosmoLattice`'s default measurement tools, we extract average values of the field to study its behavior and confirm the formation of domain walls. Additionally, we record the potential energy at each lattice site, which we then use to visualize the domain walls and qualitatively verify that they enter a scaling regime. See Appendix B for the model definition and its parameter file.

2.3 Simulating global strings

To accurately simulate cosmic strings governed by the potential in Eq. (1.18), we follow the same considerations as in the domain wall case. Comparing the respective widths and masses, we find that they are the same for strings and domain walls, so the conditions established previously still apply here. Specifically, we run our simulations in conformal time η , with parameters $v = 1$, $\lambda = 1/2$, and lattice size $N = 256$. We choose $L = 38$, which is larger than what the optimal condition in Eq. (2.7) suggests. This is necessary to ensure a sufficient number of Hubble patches at the time of formation. While this setup favors condition (2.6), it goes against (2.5) such that $\eta_{\max} \approx 8.5$ in radiation domination. We note that the complex field takes the form $\varphi = (\varphi_1 + i\varphi_2)/\sqrt{2}$, so that when $|\varphi|$ is close to unity, its real and

imaginary components follow suit. The choice of initial conditions will be the same as in the previous case.

Apart from the average field values and the potential energy, we will also examine other quantities of interest. To visualize the strings, we will plot in 3D the potential energy isosurfaces which allows the strings to be revealed. We will also plot the phase of the field to verify that, at the locations of strings, the phase indeed winds through all values in the range $[0, 2\pi)$, as expected from the theory. To enable this, the source code has been slightly modified to extract the field values at every spacetime point in the simulation box, a feature not available by default. For a detailed explanation of the modifications, see Appendix B.2. Finally, we will study the evolution of the power spectrum of the field. The power spectrum encodes the variance per unit k -volume of the field fluctuations at a given scale k , where k is the magnitude of the wavevector in the spatial Fourier transform of the field. This wavenumber is related to the wavelength of the corresponding oscillation by $\lambda = 2\pi/k$. See Appendix A for the precise definitions related to the power spectrum. *CosmoLattice* outputs by default the dimensionless power spectrum $\Delta^2(\vec{k})$ of the simulated fields. We have to take into account that the lattice (or rather its Fourier transform) only allows modes within a finite range of wavenumbers, defined by the infrared and ultraviolet cutoffs k_{IR} and k_{UV} . The infrared cutoff arises because the periodic boundary conditions restrict the longest possible wavelength to the size of the box, yielding $k_{\text{IR}} = 2\pi/L$. On the other hand, the lattice resolution limits the smallest possible wavelength in each direction $k_{i,\text{UV}} = Nk_{\text{IR}}/2$ yielding the maximum possible scale $k_{\text{UV}} = \sqrt{3}Nk_{\text{IR}}/2$. When initializing the simulation, a cutoff scale k_{cutoff} can also be specified to restrict the range of initial fluctuations to modes with $k < k_{\text{cutoff}}$. In our case, we choose this value such that the corresponding wavelength is of order unity, motivated by the fact that the Hubble parameter is normalized to unity at the initial time and causality has to be respected.

3 Simulation results

Here we present the results of the domain wall and global string simulations. We analyze the formation and evolution of the defects and compare our findings with the theoretical predictions and related studies.

3.1 Domain walls

In Fig. 3.1, we show how the average value of the field, $\langle\phi\rangle$, and its RMS, σ_ϕ , evolve during the simulation. The average stays close to zero for most of the time, only drifting a bit at the end. What's more interesting is the behavior of σ_ϕ . It starts off around 0.5, which just reflects the initial random fluctuations. Then, for $0 \leq \eta \leq 3$, it drops as the field starts settling at $\phi = 0$. At $\eta \approx 3$ there's a sharp jump to about 1, marking the moment the field rolls down to either $-v$ or v in different regions and domain walls form. At late times, the number of Hubble patches in the box gets smaller as the universe expands, leading to the domination of the $v = -1$ vacuum.

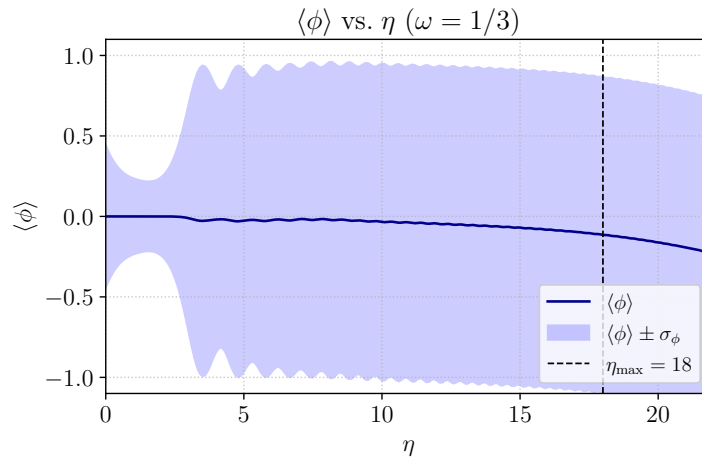


Figure 3.1: Spatially averaged field value $\langle\phi\rangle$ over the box and its RMS σ_ϕ as a function of conformal time η in a radiation-dominated background with $L = 19$ and $N = 256$ under the $\mathbb{Z}_2$ potential with $\lambda = 1/2$ and $v = 1$.

The previous plot is complemented with 2D snapshots of the value of the potential in the box, shown in Fig. 3.2. We have also drawn a dashed circle representing the size of a Hubble patch, which should contain around a constant amount of domain walls at every time if a scaling regime is achieved. We see that at early times the field is still randomly distributed, up until around $\eta = 3$, where wall-like structures begin to appear. At later times, the circle does indeed contain about one domain wall. Additionally, between the third and fourth snapshots, a wall interaction is observed where two roughly parallel walls collide and a segment breaks off. In the final snapshot, we also see how the walls have been thinned out, and there is about a single Hubble patch in the box. Finally, it's worth noting that the two walls seen in the fourth snapshot are actually the same wall, because of the periodic boundary conditions. This doesn't accurately represent real dynamics and further illustrates the need for a large number of Hubble patches in the simulation volume. With the help of Fig. 3.1 we deduce that the region «outside» the wall at late times must have the field at $\phi = -1$ given that it is the larger of the two. The small ripples in energy in regions where the field has settled into a particular vacuum correspond to scalar radiation.

In the case of matter domination, we have observed similar behavior in both the formation and evolution of the field. It is important to keep in mind that in this case the background scales faster. See Appendix C for the full results.

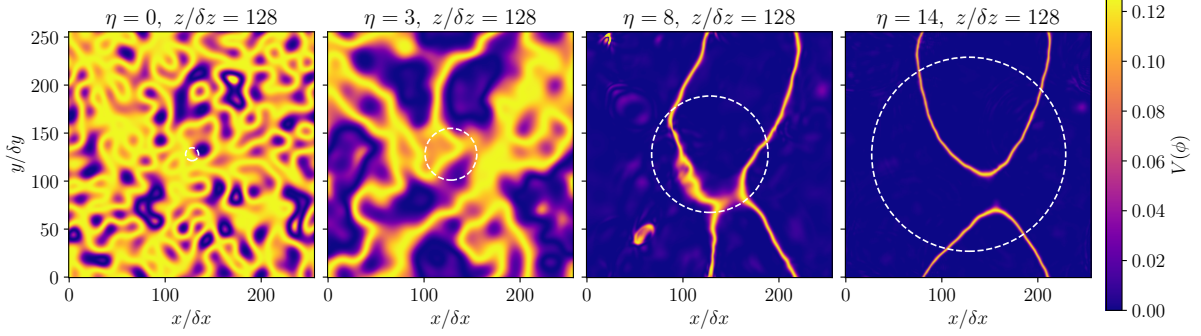


Figure 3.2: 2D snapshots at different values in conformal time η of the potential energy density in the box at a fixed height $z = L/2$ for a radiation-dominated ($\omega = 1/3$) background with $L = 19$, $N = 256$, $\lambda = 1/2$ and $v = 1$. The dashed circle in white has diameter of a Hubble radius representing a Hubble patch. See Appendix C.1 for the full evolution.

3.2 Cosmic strings

In Fig. 3.3, we show several quantities of the complex field averaged over the simulation box as a function of conformal time. Initially, the modulus of φ stays small and positive until about $\eta \approx 3$, when cosmic strings begin to form and $|\varphi|$ approaches unity, matching the radius of the circle of minima of the potential. At the same time, the real and imaginary parts of the field fluctuate similarly to the domain wall case, indicating that the field remains uniformly distributed along the circle of minima. Unlike what we observe in Fig. 3.1, there is no clear drift toward a specific minimum at late times; this is because the chosen box size L is larger than the optimal conditions suggest, increasing the number of Hubble patches present at $\eta_{\max}$ and resulting in a more uniform field distribution by the end of the simulation.

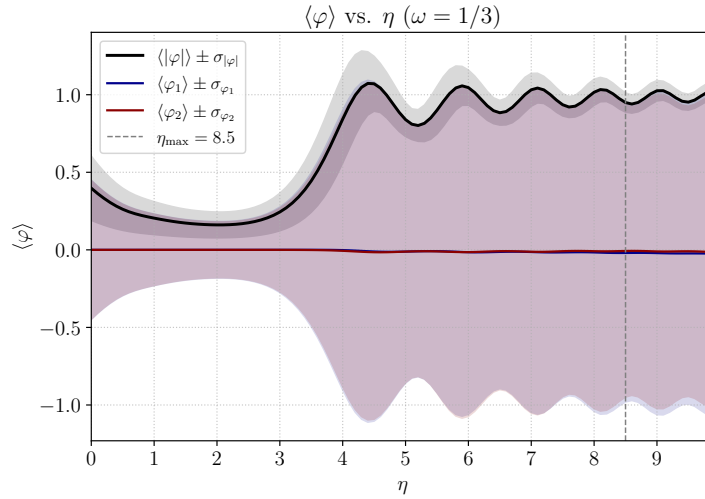


Figure 3.3: Spatially averaged value of the norm $|\varphi|$ of the complex field $\varphi = (\varphi_1 + \varphi_2)/\sqrt{2}$ and its components as a function of conformal time η in a radiation-dominated background with $L = 38$ and $N = 256$ under the Mexican hat potential with $\lambda = 1/2$ and $v = 1$.

In Fig. 3.4, we show the phase and potential energy of a two-dimensional slice through the simulation box, obtained with the mentioned modifications to the source code (Appendix B.2). As in the case of domain walls, the hot spots in the potential energy correspond to global strings, which are not always fully visible in a 2D slice. However, by comparing both plots, we can confirm that at the locations of the strings, the phase of the field spans all possible values, indicating a full winding around the vacuum manifold. It should be noted that this pattern may not be observed if a segment of a string lies parallel

to the slice, though such configurations are rare. The aspect of the phase plot closely resembles that obtained in the string-wall case in [HKS11].

$\text{Arg}(\varphi)$ and $V(\varphi)$ at $\eta = 7$, $z/\delta z = 128$ ($\omega = 1/3$)

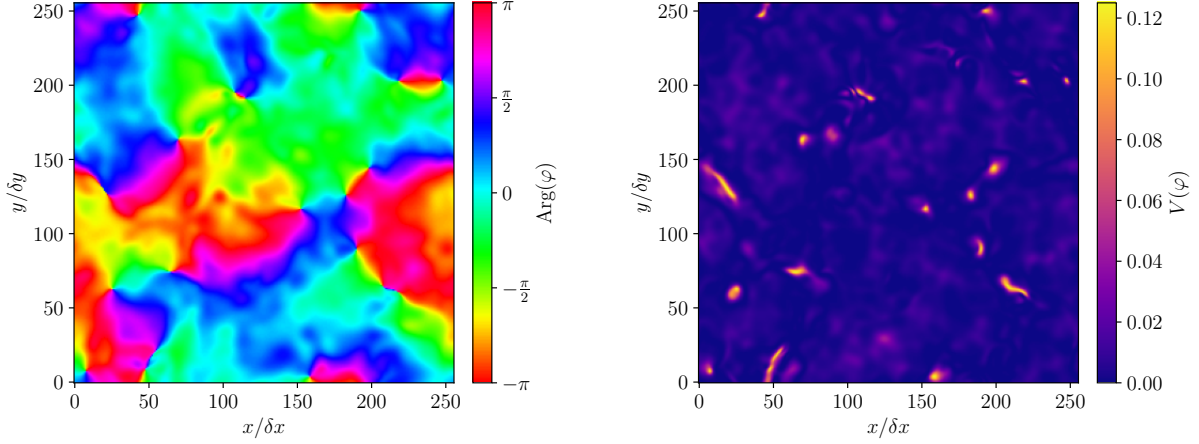


Figure 3.4: 2D snapshot at time $\eta = 7$ of the argument of the complex field φ and its potential energy in the box at height $z = L/2$ for a radiation dominated background ($\omega = 1/3$) with $L = 38$, $N = 256$, $\lambda = 1/2$ and $v = 1$.

In Fig. 3.5, we display isosurfaces of the potential energy at a value close to its maximum, $V(0) = 0.125$, taken at two different times. As expected, irregular string-like structures form and become visible. Some qualitative trends are evident: the string network becomes sparser over time, and the individual strings appear thinner. Additionally, we observe closed loops of varying sizes, some of which are in the final stages of annihilation.

Potential isosurfaces at different times

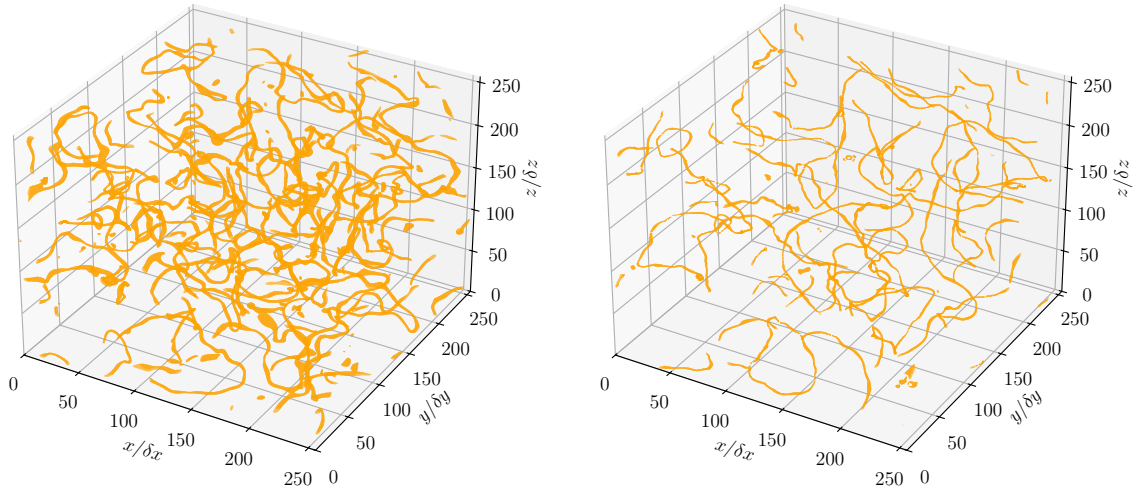


Figure 3.5: 3D snapshots at $\eta = 6, 9$ of the potential energy isosurfaces $V(\varphi) = 0.9$ after global strings have formed for a radiation dominated background ($\omega = 1/3$) with $L = 38$, $N = 256$, $\lambda = 1/2$ and $v = 1$.

If a network of topological defects is in a scaling regime we expect the power spectrum to evolve in a self-similar manner. In particular, the peak corresponding to the characteristic scale k at which most oscillations are concentrated should remain stationary in physical units. In Fig. 3.6, we plot the power

spectrum of $|\varphi|$ for $0 \leq \eta \leq 9$. The slight shift of the beginning of each spectrum with increasing time is due to the fixed k_{IR} , which appears time-dependent as a function of $k/(2\pi\mathcal{H})$. At early times, the power spectrum displays an abrupt drop at $k \approx 0.6$, which arises from the explicit cutoff in k introduced in the initial conditions, as discussed in §2.3. As time progresses, the spectrum evolves towards a stable shape, which it begins to approach around $\eta \approx 5$, coinciding with the formation of the string network. After string formation, the spectrum becomes nearly stationary, indicating that the network has entered a scaling regime. The resulting shape grows up to a peak scale k_{peak} , and then falls off again. Using data from scaling times $6 \leq \eta \leq 9$, we have fitted a parabola to the region near the maximum to identify the location of the peak. We find $k_{\text{peak}} = 0.53 \pm 0.01$, with the small uncertainty confirming the peak's stationarity. Notably, this scale is close to the initial cutoff, suggesting that the dominant scale in the system stays roughly the same throughout the evolution.

At each side of the peak we have fitted a power-law to measure the rate at which oscillations fall off. Despite the limited amount of data points in the IR region, we find that $\Delta_{|\varphi|}^2(\vec{k}) \propto (k/\mathcal{H})^3$ which is expected given that the relation with the physical power spectrum $\Delta_{|\varphi|}^2(\vec{k}) \propto k^3 P_\varphi(\vec{k})$. Indeed, this means that at IR scales $P_\varphi(\vec{k})$ stays roughly constant which is what we expect from causality constraints at larger scales, since this implies $\mathcal{H} \propto k = 2\pi/\lambda$ with λ the comoving wavelength. The fit in the UV region is done up to a certain threshold of $k/(2\pi\mathcal{H}) \approx 3.5$, right where the spectrum begins to develop a knee caused by the lack of available time for the smaller scales to develop. In this region we find that $\Delta_{|\varphi|}^2(\vec{k}) \propto (k/\mathcal{H})^{-1.66}$. We draw a rough comparison with Figure 13 in [GHV21] which shows the energy density instead of the power spectrum, however this should display a similar behavior and indeed the shapes resemble each other and the magnitude of the slope in IR is slightly larger than in UV in both.

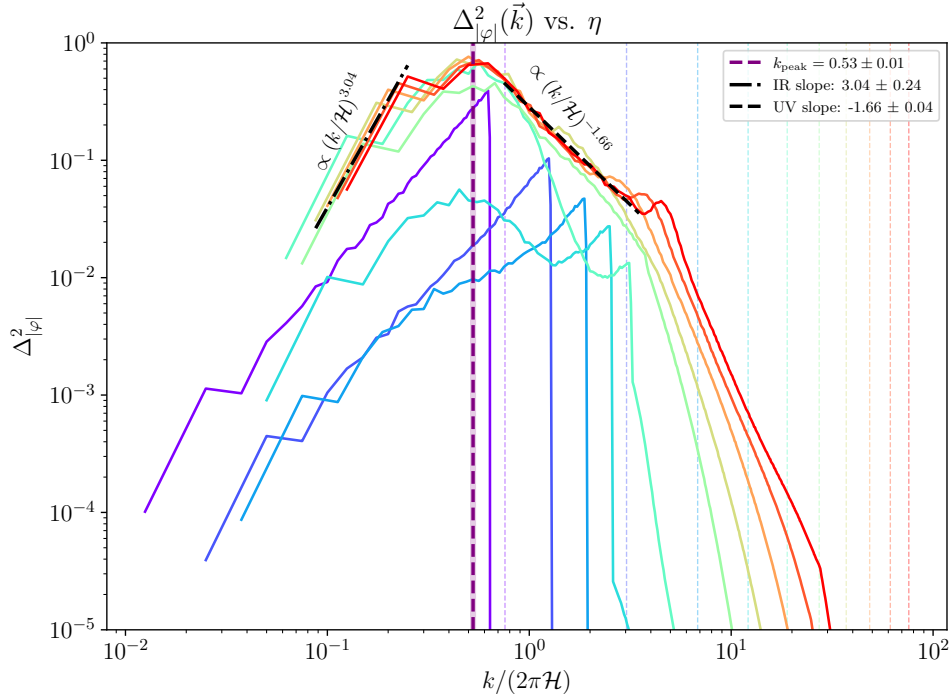


Figure 3.6: Dimensionless power spectrum of $|\varphi|$ vs. $k/2\pi\mathcal{H}$ for values of η between 0 (purple) and 9 (red). The low opacity dashed vertical lines represent the UV cutoff at the time corresponding to their color. The purple dashed line represents the fitted maximum of the distribution for values of $\eta \geq 6$. The black slopes indicated linear fit to a power law at each side of the peak (IR and UV). The data comes from a global string simulation in a radiation dominated background ($\omega = 1/3$) with $L = 80$, $N = 440$, $k_{\text{cutoff}} = 4$, $\lambda = 1/2$ and $v = 1$.

4 Conclusion

In the first half of this thesis, we established the necessary foundations in cosmology and classical field theory to provide a technically accurate description of the conditions required for defect formation, emphasizing the role of SSB and expansion. We examined how the nature of the broken symmetry determines the geometry of the resulting topological defects, and how the absence of causal contact between regions of the early universe plays a central role in their formation. We then focused on domain walls, which are two-dimensional objects in three-dimensional space that separate regions associated with different vacua, arising from real scalar field theories with discrete minima. We described their scaling behavior and explained why their energy density evolution suggests incompatibilities with the present universe. Following this, we examined global cosmic strings, viewed as one-dimensional analogs of domain walls, and discussed their formation, basic properties, and scaling behavior. For both types of defects, we briefly reviewed current research into their cosmological relevance, particularly their possible roles as sources of gravitational waves or dark matter.

In the second half, we examined the numerical techniques used to probe the dynamics of topological defects in expanding backgrounds. For this, we employed the open software *CosmoLattice* to set up and run the models introduced in the first half. We successfully simulated and visualized both domain walls and strings, analyzing the behavior of their scalar fields and the scaling of the defects. In the case of strings, we were able to modify the source code to plot the field phase and we also looked at their power spectrum, concluding that $\Delta_{|\varphi|}^2 \sim (k/\mathcal{H})^{3.04}$ in IR as expected from causality and $\Delta_{|\varphi|}^2 \sim (k/\mathcal{H})^{-1.66}$ in UV. Throughout this process, a solid understanding of the software was developed, along with a clear view of the impact of various numerical and physical parameters on accurately capturing defect dynamics.

Finally, we conclude that both objectives of this project were achieved in a satisfactory manner, and we suggest several possible directions for future research. A natural next step, which barely escaped the scope of this work, is the study of hybrid defects arising from theories with sequential symmetry breaking. Particularly relevant to this thesis are domain walls bounded by strings, which appear in models where a $U(1)$ symmetry is broken followed by a $\mathbb{Z}_2$ (or $\mathbb{Z}_n$) symmetry [VS00]. Such defects arise naturally in axion models which have already been constrained by numerical studies [HKS11]. Another suggested direction is the study of other types of defects, such as local cosmic strings or monopoles, which are point-like. In all these cases, the formation process, energy scaling, and emission of particles or gravitational waves are of great interest in the current cosmological landscape, especially as their detection motivates ongoing and future missions, including GW observatories and CMB experiments.

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A The power spectrum of a field

The power spectrum is a very useful tool for analyzing a field's behavior in terms of its *Fourier modes*, especially when the field exhibits oscillatory behavior. If the field is statistically homogeneous and isotropic, its statistical properties can be described entirely by the two-point correlation function, which is diagonal in Fourier space. With the time fixed, the *spatial Fourier transform* of a scalar field $\phi(\vec{x})$ is defined as

$$\phi(\vec{k}) = \int d^3x \phi(\vec{x}) e^{-i\vec{k}\cdot\vec{x}}, \quad (\text{A.1})$$

where $\vec{k}$ is the *wavevector*, pointing in the direction of oscillation and its norm $k = \|\vec{k}\|$ is related to the *wavelength* by $\lambda = 2\pi/k$. We then define the *power spectrum* $P_\phi(\vec{k})$ as the amplitude of the fluctuations at scale k . Since the field is homogeneous and isotropic we assume the different scales are uncorrelated

$$\langle \phi(\vec{k}) \phi(\vec{k}') \rangle = (2\pi)^3 \delta^3(\vec{k} - \vec{k}') P_\phi(\vec{k}) \quad (\text{A.2})$$

where $\langle \phi(\vec{k}) \phi(\vec{k}') \rangle$ is the correlation of the field at different scales. Thus, $P_\phi(k)$ encodes the *variance per unit k -volume* of the field fluctuations at scale k and as a consequence

$$\langle \phi(\vec{x})^2 \rangle = \int \frac{d^3k}{(2\pi)^3} P_\phi(\vec{k}). \quad (\text{A.3})$$

It is often more useful to work with the *dimensionless power spectrum*, defined as

$$\Delta_\phi^2(\vec{k}) \equiv \frac{k^3}{2\pi^2} P_\phi(\vec{k}), \quad \text{such that} \quad \langle \phi(\vec{x})^2 \rangle = \int d \ln k \Delta_\phi^2(\vec{k}); \quad (\text{A.4})$$

given that $d^3k = 4\pi k^2 dk$. So $\Delta_\phi^2(\vec{k})$ quantifies the contribution to the variance of the scale k per logarithmic interval in k .

B Use of *CosmoLattice*

Here we detail how we have used the *CosmoLattice* software to obtain the results in this thesis.

B.1 Example of a model and a parameter file

To define a model in *CosmoLattice*, the user must provide both a model file and a corresponding parameter file. In the model file, the choice of time coordinate is specified by assigning a value to the variable α , where $\alpha = 0$ corresponds to cosmic time and $\alpha = 1$ to conformal time. The user must also define the parameters of the potential as variables and assign values to the dimensionless quantities f_* and ω_* appearing in Eq. (2.2). In addition, it is necessary to specify the number and type of fields involved, as well as to provide explicit expressions for the potential energy and its first and second derivatives.

In Listing 1 we give the code snippets of the model file used to simulate domain walls in radiation domination. The case of matter domination is the same with $\alpha = 0$.

```

1 struct ModelPars : public TempLat::DefaultModelPars {
2     static constexpr size_t NScalars = 1;
3     static constexpr size_t NPotTerms = 1;
4 };
5
6 ...
7
8 private:
9
10 double lambda, v;
11
12 ...
13
14 alpha = 1;
15 fStar = 1;
16 omegaStar = fStar;
17
18 ...
19
20 auto potentialTerms(Tag<0>)
21 {
22     return 0.25 * lambda * pow<4>(fldS(0_c))
23         - 0.5 * lambda * pow<2>(fldS(0_c)) * v
24         + 0.25 * lambda * pow<4>(v);
25 }
26
27 ...
28
29 auto potDeriv(Tag<0>)
30 {
31     return lambda * pow<3>(fldS(0_c)) - lambda * fldS(0_c);
32 }
33
34 ...
35

```

```

36 auto potDeriv2(Tag<0>)<...>
37 {
38     return 3 * lambda * pow<2>(fldS(0_c)) - lambda;
39 }

```

Listing 1: Simplified *CosmoLattice* model definition file for a domain wall simulation. Ellipses "*...*" indicate omitted code.

The text below corresponds to the precise parameter file (in *.txt* format) inputted to the model.

```

#Output
outputfile = ./

#Evolution
expansion = true
evolver = WV2
fixedBackground = true
omegaEoS = 0

H0 = 1

#Lattice
N = 256
dt = 0.02
lSide = 19

#Times
tOutputFreq = 0.1
tOutputInfreq = 1
tMax = 22

tOutputRareFreq = 1
energy_snapshot = E_V

#Spectra options
PS_type = 1
PS_version = 1

#GWs
GWprojectorType = 1
withGWs = false

#IC
kCutOff = 4
initial_amplitudes = 0.0
initial_momenta = 0.0

#Model Parameters
lambda = 0.5
v = 1.0

```

B.2 Modification to the measurements

Most of the quantities output by *CosmoLattice* are spatially averaged or otherwise processed, since storing the raw field data at every lattice site is typically very memory-intensive. However, the program provides an option to save the kinetic, gradient, and potential energy densities as binary `.hdf5` files. We exploited the fact that the code allows explicit separation of the individual terms composing the potential to isolate and save the values of the fields themselves. To achieve this, we add and subtract terms within the potential so that its overall form remains unchanged, but now includes isolated contributions such as the real part of a scalar field. This enables *CosmoLattice* to record these specific field components by modifying the portion of the code responsible for the box measurements. In particular, we used this method to extract the real and imaginary components of the field in §3.2 to plot the phase of the field.

In Listing 2, we show how the real and imaginary parts of the complex field φ are added and subtracted in the potential so that they appear as separate terms without modifying the overall form of the potential. This allows us to isolate specific field components. To record these terms individually, we modify the program file

```
cosmolattice/src/include/CosmoInterface/measurements/energysnapshotmeasurer.h
```

which handles the output of `.hdf5` files. In particular, we extend it to allow printing of individual potential terms—see Listing 3. Finally, we modify

```
cosmolattice/src/include/CosmoInterface/definitions/potential.h
```

by adding functions that return only the values of a specific term in the potential rather than the sum of all terms (order of the terms matters)—see Listing 4.

```

1 struct ModelPars : public TempLat::DefaultModelPars{
2     ...
3     static constexpr size_t NCSalars = 1;
4     ...
5     static constexpr size_t NPotTerms = 5;
6 };
7
8 ...
9
10 auto potentialTerms(Tag<0>)
11 {
12     return lambda*0.25*(pow<4>(norm(fldCS(0_c)))-2*pow<2>(norm(fldCS(0
13         _c)))*pow<2>(v)+pow<4>(v));
14 }
15
16 auto potentialTerms(Tag<1>) //
17 {
18     return Real(fldCS(0_c));
19 }
20
21 auto potentialTerms(Tag<2>) //
22 {
23     return -Real(fldCS(0_c));
24 }
25
26 auto potentialTerms(Tag<3>) //
```

```

26     {
27         return  Imag(fldCS(0_c));
28     }
29
30     auto potentialTerms(Tag<4>) //
31     {
32         return  -Imag(fldCS(0_c));
33     }

```

Listing 2: *CosmoLattice* model potential definition file for a global string simulation. Ellipses "..." indicate omitted code.

```

1  template <typename Model>
2      EnergySnapshotsMeasurer(Model& model, FileManager& fm, std::vector<std
      ::string> toSave) :
3      mRoot(fm.getTag()+fm.getWorkingDir())
4      {
5          // This checks which energies are specified in the string "toSave"
           (passed as a parameter), and creates the corresponding h5 files
           to save the snapshots.
6
7          ...
8
9          saveFieldComps =  IsInContainer::check("Comps", toSave); // save
           the field components in 2 hdf5 files
10
11      ...
12
13      // if the option to print the field components is selected create the .
           hdf5 files
14      if(saveFieldComps) {
15          nameFieldReal = mRoot + "real_field_component.h5"; // name of
           the file
16          fIO.saver.create( nameFieldReal );
17          fIO.saver.close();
18
19          nameFieldComplex = mRoot + "complex_field_component.h5"; //
           name of the file
20          fIO.saver.create( nameFieldComplex );
21          fIO.saver.close();
22      }
23
24      ...
25
26      // if the option to print the field components is selected call the
           function that prints them
27      if(saveFieldComps) {
28          fIO.saver.open( nameFieldReal );
29          fIO.saver.save(t, Model::fieldReal(model), "field_real");
30          fIO.saver.close();
31
32          fIO.saver.open( nameFieldComplex );

```

```

33         fIO.saver.save(t, Model::fieldComplex(model), "field_complex");
34         fIO.saver.close();
35     }
36
37 ...
38
39 private:
40     /* Put all member variables and private methods here. These may change
41        arbitrarily. */
42     FileIO fIO;
43     std::string mRoot;
44
45     ...
46
47     bool saveFieldComps;
48
49     ...
50
51     std::string nameFieldReal, nameFieldComplex;

```

Listing 3: *CosmoLattice* modification to the `energysnapshotmeasurer.h` file to add options of printing individual potential terms. Ellipses `"..."` indicate ommited code.

```

1
2     class Potential {
3     ...
4
5     public:
6
7         ...
8
9         template<class Model>
10        static inline auto fieldReal(Model& pM)
11        {
12            return Total(i,1,1,pM.potentialTerms(i));
13        }
14
15        template<class Model>
16        static inline auto fieldComplex(Model& pM)
17        {
18            return Total(i,3,3,pM.potentialTerms(i));
19        }

```

Listing 4: *CosmoLattice* modification to the `potential.h` file to add options of printing individual potential terms. Ellipses `"..."` indicate ommited code.

C Gallery of additional results

C.1 Domain walls in radiation domination

We include the full evolution of the 2D snapshots of the radiation-dominated DW simulation in §3.1.

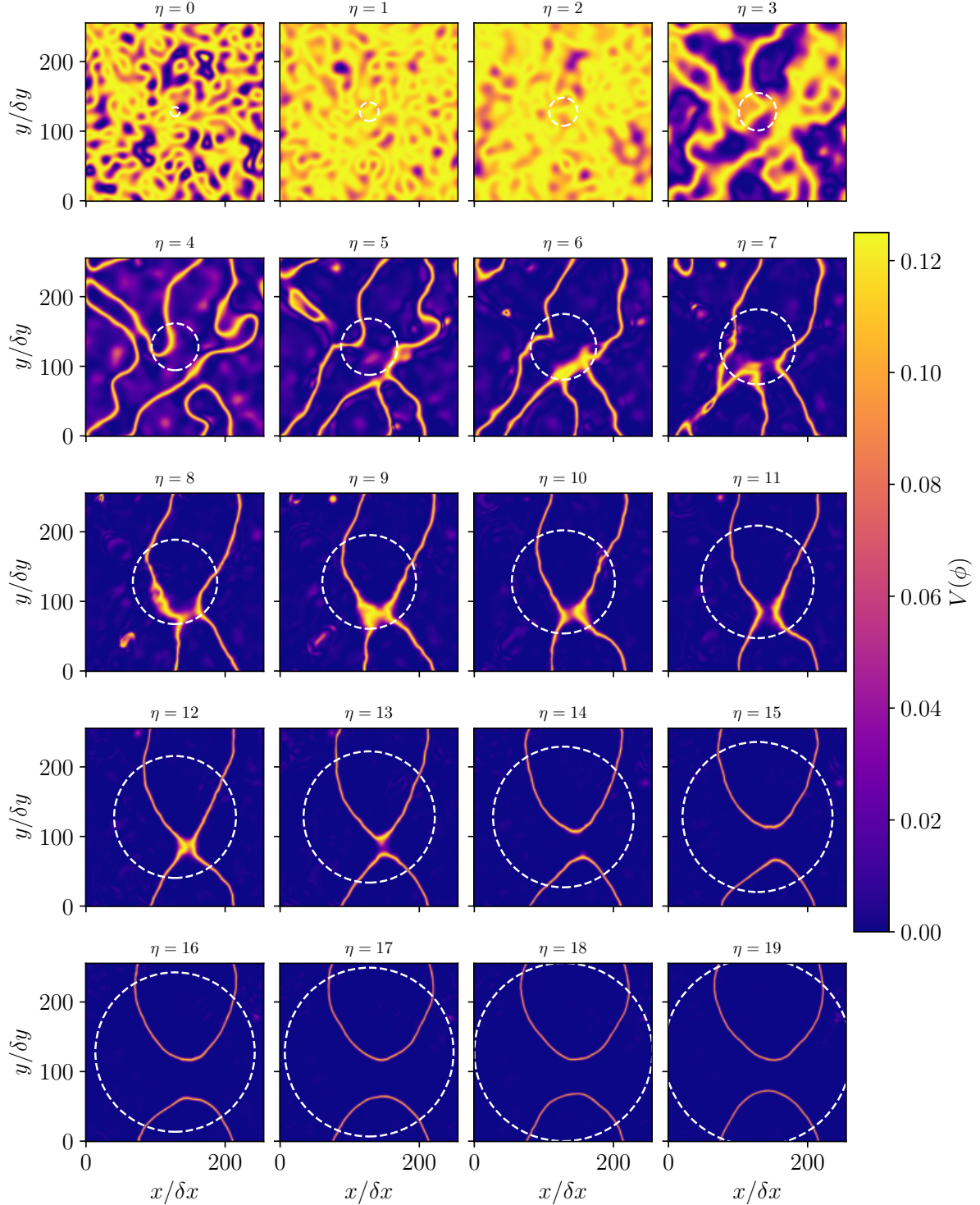


Figure C.1: Snapshots at different values in conformal time η of the potential energy density in the box at a fixed height $z = L/2$ for a radiation-dominated background ($\omega = 1/3$) with $L = 19$, $N = 256$, $\lambda = 1/2$ and $v = 1$. The dashed circle in white has diameter of a Hubble radius thus representing a Hubble patch.

C.2 Domain walls in matter domination

For matter domination ($p = 2$), the optimal condition (Eq. (2.7)) yields $L \approx 7$, which is too small to ensure sufficient Hubble volume coverage—especially considering that domain wall formation is not instantaneous and expansion is faster than in radiation domination. To improve comparability and resolution, we relax the optimal condition and choose $L = 19$. This reduces the maximum time to $\eta_{\max} \approx 6.7$ in conformal time. To better temporally resolve the domain wall dynamics, it is therefore convenient to switch to cosmic time, where the corresponding maximum time is significantly longer $t_{\max} \approx 54$. The conversions between t and η are given by

$$t = \frac{p}{p+1} \left[\left(1 + \frac{\eta}{p} \right)^{p+1} - 1 \right], \quad (\text{C.1})$$

which gives the scale factor and Hubble parameter in terms of t as

$$a(t) = \left(1 + \frac{p+1}{p} t \right)^{\frac{p}{p+1}}, \quad H(t) = \frac{1}{1 + \frac{p+1}{p} t}, \quad (\text{C.2})$$

where again we have assumed initial normalization conditions $a(t=0) = 1$, $H(t=0) = 1$.

In Fig. C.2, we observe that the main difference with radiation domination is that, since cosmic time progresses more slowly than conformal time, the dynamics of the field appear slower. We also note that, as described previously, the walls become too thin to resolve before the number of Hubble patches in the box drops significantly. In Fig. C.3 we show the initial and final parts of the evolution of a 2D potential energy snapshot corresponding to the simulation of Fig. C.2.

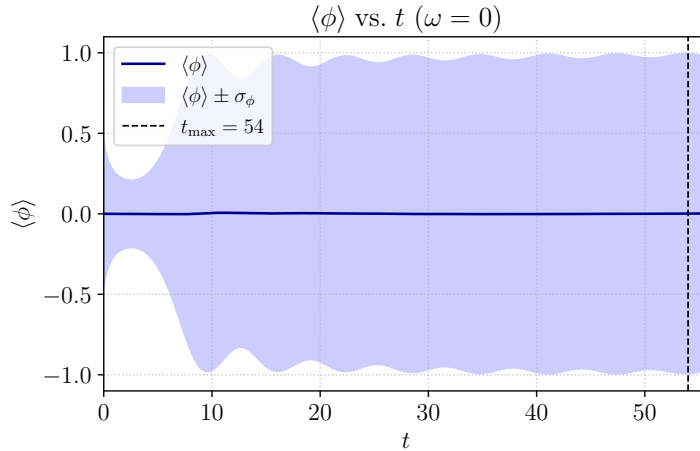


Figure C.2: Spatially averaged field value $\langle \phi \rangle$ over the box and its RMS σ_ϕ as a function of cosmic time t in a matter-dominated background with $L = 19$ and $N = 256$ under the $\mathbb{Z}_2$ potential with $\lambda = 1/2$ and $v = 1$.

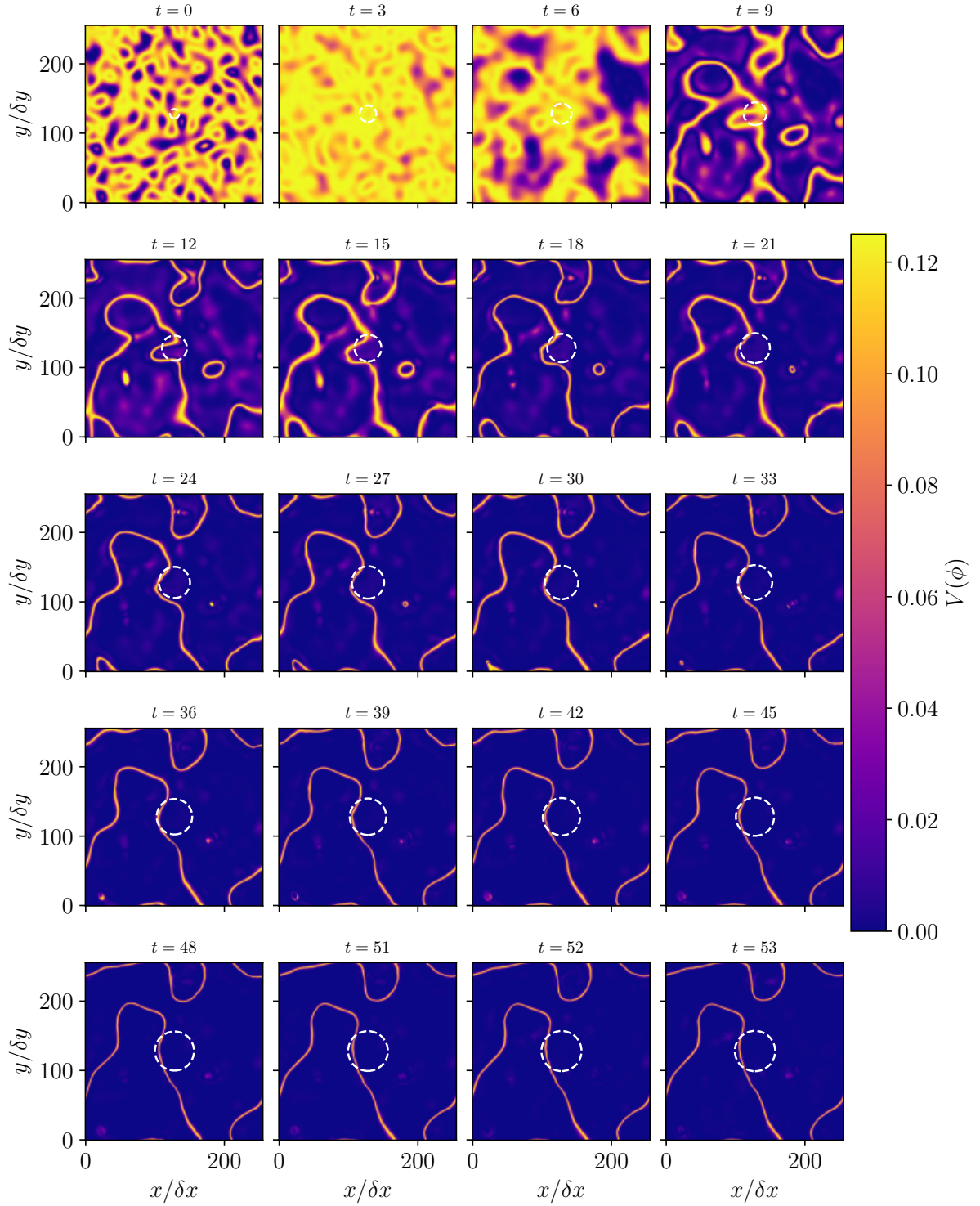


Figure C.3: Snapshots at different values in cosmic time t of the potential energy density in the box at a fixed height $z = L/2$ for a matter-dominated background ($\omega = 0$) with $L = 19$, $N = 256$, $\lambda = 1/2$ and $v = 1$. The dashed circle in white has diameter of a Hubble radius thus representing a Hubble patch.